

Muhammad Shiraz Kamran

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Education

COMSATS University Islamabad

Islamabad, Pakistan

Bachelor of Science in Software Engineering (CGPA: 3.62 / 4.0)

Expected 2027

- **Relevant Courses:** Database Systems, Object-Oriented Programming (OOP), Data Structures Algorithms, Design and Analysis of Algorithms, Computer Networks, Programming Fundamentals (PF), Software Requirements Engineering, Software Design and Architecture.

Technical Skills

Languages	Python, Java, JavaScript, SQL (PostgreSQL, sqlite), HTML/CSS
Backend	FastAPI, Spring Boot, Node.js, Fastify
AI & Data	Supabase, Postgresql, sqlite, Sentence Transformers (NLP), Vector Embeddings
Cloud	VPS handling, DigitalOcean, AWS training intro, Netlify, Supabase
Tools & DevOps	Git, Docker, Tampermonkey, Netlify, Jira, Postman
Frontend	React.js, React Router, Tailwind CSS
Certificates & Courses	Cisco Python Essentials 2, Intro to AWS Cloud with Labs

Projects

Hadith Semantic Search System (RAG) | *Python, FastAPI, React, Qdrant, Docker* [Github](#) · [Live Demo](#)

- Developed a Retrieval Augmented Generation(RAG) semantic search system for **Sahih Bukhari**, enabling meaning based search.
- Used **BAAI/bge-small-en-v1.5** embeddings with **Qdrant Cloud** as vector database for similarity search.
- Added validation and summaries using **Gemini API** with **Groq API** as fallback.
- Built with **FastAPI** and **React**, deploys using CI/CD with **GitHub Actions** and **Docker Compose** on **VPS** at Digital Ocean.

Filebin CLI - Temporary File Sharing | *Python, Fastify (Node.js), Supabase* [GitHub](#) · [PyPI](#)

- Developed a CLI tool for temporary file sharing using the Python **click** library and published it on **PyPI** with support for Python 3.7+.
- Built a backend using **Fastify** (Node.js) and **Supabase** to manage file metadata efficiently without needing to self-host the files.
- Implemented global rate-limiting to handle up to **4,000 requests per minute** to ensure the API remains stable under heavy traffic.

Munch n Vibes - Bakery E-Commerce | *React.js, React Router, CSS3, Netlify* [Live Site](#)

- Developed a multi-page **React.js** website for a home baker using a glassmorphism UI and **React Router** for smooth navigation and dynamic product routing.
- Implemented responsive layouts and deployed the application via **Netlify** and **GitHub** to handle continuous integration and automatic updates.

Devotol - AI Developer Assistant | *JavaScript, Gemini 2.0 Flash, Netlify* [Live Site](#)

- Built a lightweight AI assistant using **JavaScript** with a bloat free, minimal architecture to ensure fast load times and quick responses for coding prompts.
- Developed a serverless backend layer using **Netlify Functions** to securely integrate and manage requests to the **Google Gemini 2.0 Flash** model.

Volunteering

Gaza Relief Fundraising Campaign

Team Co-Lead

- Co-led a team of 5 students to execute a fundraising campaign for Civics and Community Engagement, successfully raising over **350,000 PKR** through community outreach and donation drives.